

Agenda

1. Announcements

- Midterm 2 is next Tuesday April 11 at 7:30pm

Exam Info Page: go.wisc.edu/1xxzfx

- Please fill out your TA evaluations

2. Section 11.9 - Representing Functions as Power Series

Representing Functions as Power Series

Recall from Calculus I: $\frac{d}{dx}[x^n] = nx^{n-1}$ and $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Theorem: Suppose we have the power series $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$ with radius of convergence R . Then f is differentiable and integrable on the domain $(c-R, c+R)$ where:

1.) $f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1} = a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + 4a_4(x-c)^3 + \dots$

2.) $\int f(x) dx = C + \sum_{n=0}^{\infty} a_n \frac{(x-c)^{n+1}}{n+1} = C + a_0(x-c) + a_1 \frac{(x-c)^2}{2} + a_2 \frac{(x-c)^3}{3} + \dots$

Note: $f(x)$, $f'(x)$, and $\int f(x) dx$ all have the same radii of convergence (but the intervals of convergence can differ)
because of the endpoints

Activity: $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$ has ROC: 1
IOC: $[-1, 1)$

1.) Find the IOC of $f'(x)$

$$\sum_{n=1}^{\infty} \frac{n x^{n-1}}{n} = \sum_{n=1}^{\infty} x^{n-1}$$

2.) Find the IOC for $\int f(x) dx$

① Find $f'(x)$
② Test endpoints
converges with $|x| < 1$
 $(-1, 1)$

$$\begin{aligned} \int f(x) dx &= C + \int \left[\sum_{n=1}^{\infty} \frac{x^n}{n} \right] dx \\ &= C + \int \frac{1}{n} \left(\frac{x^{n+1}}{n+1} \right) + C \\ &= C + \sum_{n=1}^{\infty} \frac{x^{n+1}}{(n+1)n} \end{aligned}$$

① Find $\int f(x) dx$
② test endpoints

does $\sum \frac{1}{n^2+1}$ converge.

yes with $0 \leq \frac{1}{n^2+1} \leq \frac{1}{n^2} \checkmark$

$\therefore$ original series is also convergent $\hookrightarrow$

Example: Find a power series representation of $[-1, 1]$ for

$y = \ln(5-x)$ and determine its radius of convergence

$$y' = \frac{-1}{5-x} = \frac{-1}{5(1-\frac{x}{5})} = \frac{-\frac{1}{5}}{1-\frac{x}{5}} = \sum_{n=0}^{\infty} \left(-\frac{1}{5}\right) \left(\frac{x}{5}\right)^n$$

converges when $\left|\frac{x}{5}\right| < 1$

$$|x| < 5$$

Radius of convergence = 5

Activity: Find a power series representation for $\frac{1}{(1-x)^2}$ and determine its radius of convergence.

$$\int \frac{1}{(1-x)^2} dx = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ converges when } |x| < 1 = R$$